

BJT Multistage Amplifiers

Theory and Design

A working note for the circuit-theory course and the companion laboratory: cascaded CE–CE and cascode stages, what cascading does to gain and bandwidth, and two worked designs you can build on a breadboard.

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Companion text: **B. Razavi**, *Fundamentals of Microelectronics*, Ch. 5, 9, 11

You already know how to bias a single common-emitter stage and analyse it with the hybrid- π model. Everything here is built on that. The new questions are: what happens when you connect two such stages end to end, why does the bandwidth get *worse* when you do, and what is the one trick (the cascode) that buys back the bandwidth without giving up the gain.

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1 Why cascade at all

A single common-emitter stage seems capable of enormous gain. With a resistive collector load and no degeneration,

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$$A_v = -g_m(R_C \parallel r_o) \quad \text{and, if } R_C \ll r_o, \quad |A_v| \approx g_m R_C = \frac{I_C R_C}{V_T} = \frac{V_{R_C}}{V_T}$$

That last form is worth memorising, because it turns the gain into something you can see on a multimeter. **The gain of a CE stage is simply the DC voltage dropped across the collector resistor, divided by 26 mV.** Nothing else. Doubling R_C or doubling I_C both work only insofar as they increase that drop.

And that is exactly where the ceiling comes from. With $V_{CC} = 12\text{ V}$ you can afford to lose maybe 5–6 V across R_C before the transistor runs out of headroom for signal swing. So $|A_v| \leq 6/0.026 \approx 230$ — and that is the optimistic, unloaded, no-degeneration number. Add a 10 k Ω load and a degeneration resistor for bias stability and you are realistically at 20–100 per stage.

Pushing further hits a second, harder wall. Letting $R_C \rightarrow \infty$ (an ideal current source load) gives the *intrinsic gain*

$$|A_v|_{\max} = g_m r_o = \frac{I_C}{V_T} \cdot \frac{V_A}{I_C} = \frac{V_A}{V_T}$$

which is independent of bias current — about 3800 for $V_A = 100\text{ V}$. You cannot exceed it with one transistor, no matter how you bias it. (Razavi makes this point in Ch. 9; it is the single most important number in analog design.)

Gain is only one of four reasons to cascade. The others are just as common in practice:

What one stage cannot do, and which stage fixes it

Requirement	Problem with a single CE stage	Usual fix
Large gain	Swing-limited to ~ 100 per stage in practice	Cascade two or more CE stages
High input resistance	$R_{in} = R_B \parallel r_\pi \approx 2\text{--}5\text{ k}\Omega$, loads the source	Put a CC (emitter follower) or a degenerated stage first
Low output resistance	$R_{out} = R_C$, kilohms — cannot drive 100 Ω	Put a CC stage last
Wide bandwidth at high gain	Miller multiplication of C_μ kills the input pole	Cascode (CE–CB)

Physical intuition

Think of each stage as a specialist rather than a generalist. A CE stage is a good *voltage multiplier* but a poor *listener* (low R_{in}) and a poor *talker* (high R_{out}). An emitter follower is the opposite: it multiplies nothing but listens and talks beautifully. A cascade lets you hire one of each instead of asking a single transistor to do all three jobs badly.

2 Types of multistage amplifiers

There are three independent ways to classify a multistage amplifier, and you should be able to answer all three for any circuit put in front of you: *how are the stages coupled*, *what configuration is each stage*, and *what job does each stage do*.

2.1 By coupling method

Coupling schemes and where each is used

Type	How	Passes DC?	Notes
RC (capacitive)	Series capacitor between collector and next base	No	Bias of each stage is independent — easy to design and debug. Costs low-frequency response. This is what your lab circuit uses.
Direct	Collector wired straight to next base	Yes	Response down to DC; no coupling capacitors. But DC levels stack up and drift/offset propagates and is amplified. Used in all ICs.
Transformer	Magnetic coupling	No	Impedance transformation and DC isolation with no resistive loss. Bulky, narrowband — RF and audio output stages.
Optical / isolated	LED–photodiode	No	Galvanic isolation for instrumentation and medical front ends.

2.2 By stage configuration

Naming is always *first stage* – *second stage*, in signal order.

Pair	What you get	Typical use
CE–CE	Very high voltage gain; non-inverting overall (two inversions); moderate R_{in} ; poor bandwidth	General-purpose gain block — Experiment 1
CE–CB (<i>cascode</i>)	Same gain as one CE stage but with the Miller effect suppressed; very high R_{out}	Wideband and high-output-impedance stages — Experiment 2
CC–CE	High R_{in} feeding a gain stage; the follower stops the source from being loaded by r_{π}	Sensor and high-impedance front ends
CE–CC	Gain first, then a buffer that can drive a low-impedance load	Output stages, headphone/line drivers
CC–CC (Darlington)	Current gain $\approx \beta_1\beta_2$; very high R_{in}	Power drivers, high-input-impedance buffers
Differential pair + CE	Common-mode rejection, then gain	Every op-amp input stage

2.3 By functional role

Any real amplifier — a 741, an audio power amp, an instrumentation front end — is organised as three functional blocks. Recognising them makes any unfamiliar schematic readable.

- **Input stage.** Optimised for input resistance, noise, offset and CMRR. Gain is secondary. Usually a differential pair or an emitter follower.
- **Gain stage.** Optimised for voltage gain alone. Usually CE or cascode, often with an active (current-source) load.
- **Output stage.** Optimised for output resistance and current delivery. Gain ≈ 1 . Emitter follower or push–pull pair.

3 Effect of cascading on gain

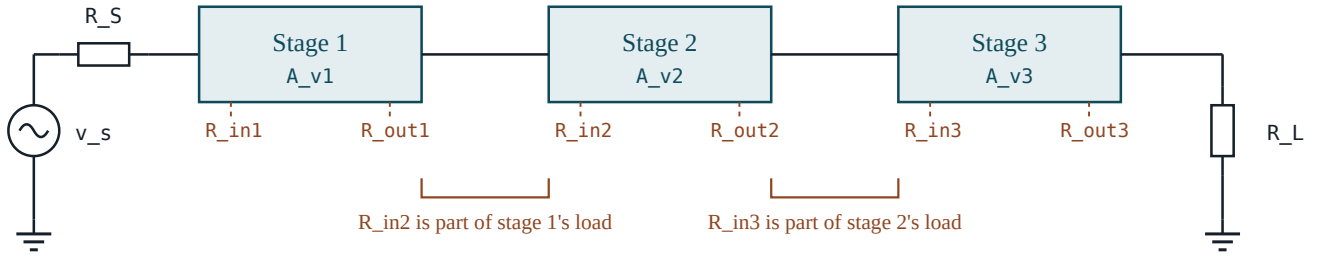


Figure 1 A cascade is a chain of loaded stages. Every stage is loaded by the *input resistance of the stage that follows it*. If you compute each stage's gain in isolation and multiply, you will overestimate the answer — often by a factor of two or more in a BJT circuit, because r_π is only a few kilohms.

The overall voltage gain is the product of the individual gains, provided each individual gain is computed *with the loading already included*:

$$A_{v,\text{total}} = \frac{v_{\text{out}}}{v_{\text{in}}} = A_{v1} \cdot A_{v2} \cdots A_{vn}, \quad A_{vk} = \frac{v_k}{v_{k-1}} \text{ loaded by } R_{in,k+1}$$

If you also want the gain referred to the signal source rather than to the amplifier's input terminals, include the input divider:

$$A_{vs} = \frac{v_{\text{out}}}{v_s} = \frac{R_{in}}{R_{in} + R_S} A_{v,\text{total}}$$

In decibels the multiplication becomes an addition, which is why frequency-response plots are drawn in dB — cascading then means literally stacking the curves:

$$A_{\text{dB}} = 20 \log_{10} |A_{v,\text{total}}| = \sum_k 20 \log_{10} |A_{vk}|$$

And the overall input and output resistances are simply those of the first and last stages:

$$R_{in} = R_{in,1}, \quad R_{out} = R_{out,n}$$

Physical intuition – the loading tax

Take two identical CE stages, each with $R_C = 4.7\text{ k}\Omega$ at $I_C = 1\text{ mA}$. On its own, stage 1 would give $|A_{v1}| = g_m R_C = 36\text{ m} \times 4.7\text{ k} \approx 169$. But the moment you connect stage 2, its $r_\pi \approx 5.5\text{ k}\Omega$ and its bias divider ($\approx 15\text{ k}\Omega$) hang across stage 1's collector. The effective load collapses to $4.7 \parallel 15 \parallel 5.5 \approx 2.2\text{ k}\Omega$ and the gain falls to about 79 — you lost more than half of it just by connecting the wire.

This is a *BJT-specific* problem. A MOSFET stage has infinite DC input resistance, so cascading MOS stages costs you nothing at midband. In a BJT cascade, the base current of the next stage is a real load, and that is the number-one reason emitter followers get inserted between gain stages.

Two practical consequences follow. First, an inverting stage flips the sign, so an even number of CE stages is non-inverting overall — your CE–CE cascade will show *no* phase inversion on the scope at midband. Second, the earlier stages should carry the *smaller* share of the gain when noise or distortion matter, because the last stage is the one handling the largest signal and is the first to clip.

4 Effect of cascading on bandwidth

Gain multiplies. Bandwidth does not — it shrinks. There are two separate mechanisms, and it matters that you keep them apart.

4.1 Mechanism A: the -3 dB point moves in even for identical, non-interacting stages

Suppose each stage is a perfect single-pole low-pass with the same corner f_H and there is no loading interaction. The magnitude responses multiply:

$$\frac{A(f)}{A_0} = \left[\frac{1}{\sqrt{1 + (f/f_H)^2}} \right]^n$$

Setting this to $1/\sqrt{2}$ and solving for the new corner gives the standard *bandwidth shrinkage factor*:

$$f_H^{(n)} = f_H \sqrt{2^{1/n} - 1}$$

$$f_L^{(n)} = \frac{f_L}{\sqrt{2^{1/n} - 1}}$$

Bandwidth shrinkage for n identical non-interacting stages

n	$\sqrt{(2^{1/n}-1)}$	Upper cut-off	Lower cut-off	Gain
1	1.000	f_H	f_L	A
2	0.644	$0.64 f_H$	$1.55 f_L$	A^2
3	0.510	$0.51 f_H$	$1.96 f_L$	A^3
4	0.435	$0.44 f_H$	$2.30 f_L$	A^4
5	0.386	$0.39 f_H$	$2.59 f_L$	A^5

Note the asymmetry, and it is good news: gain grows *geometrically* (A^n) while bandwidth shrinks only *slowly*. Two stages of gain 20 give 400 at 64 % of the bandwidth — a gain–bandwidth product 12.9× better than one stage. This is precisely why cascading is worth doing at all.

Physical intuition – why the corner moves in

At $f = f_H$ a single stage has already lost 3 dB. Two such stages in series lose 3 + 3 = 6 dB there. Since 6 dB is *past* the –3 dB definition, the overall –3 dB point must occur at a lower frequency — each stage need only lose 1.5 dB. The corner has not moved because anything physical changed inside a stage; it moved because the *definition* of the corner is now split among more stages.

4.2 Mechanism B: real cascades interact, and Miller makes it much worse

The formula above is a best case that almost never applies to a real RC-coupled BJT cascade, because the stages are not independent. The collector of Q1 and the base of Q2 are the *same node*, and that node carries:

- $C_{\mu 1}$, from stage 1's own collector–base capacitance;
- $C_{\pi 2}$, stage 2's base–emitter capacitance;
- and worst of all, $C_{\mu 2}(1 + |A_{v2}|)$ — stage 2's collector–base capacitance *multiplied by stage 2's gain* (Miller effect).

All of that sits across a resistance of a few kilohms. In the worked design of §8 this single node contributes 72 % of the total time constant. **The interstage node, not the input and not the output, is the bandwidth bottleneck of a CE–CE cascade.**

4.3 The estimation tool: open-circuit time constants

Rather than solving a fourth-order transfer function, use the method Razavi introduces in Ch. 11. For every capacitor C_i , find the resistance R_{i0} seen across its terminals with *all other capacitors open-circuited* and the independent sources zeroed. Then

$$\omega_H \approx \frac{1}{\sum_i R_{i0} C_i} \quad \Rightarrow \quad f_H \approx \frac{1}{2\pi \sum_i \tau_i}$$

This always gives a conservative (pessimistic) estimate, it is accurate to within 10–20 % for the circuits here, and — the real payoff — the list of τ_i tells you *which node to fix*. For the low end, coupling and bypass capacitors each contribute a high-pass corner and you take the dominant one (or root-sum-square them):

$$f_L \approx \sqrt{\sum_i f_{Li}^2}, \quad f_{Li} = \frac{1}{2\pi R_i C_i}$$

Watch for this in the lab

Sign convention trap: for *high*-frequency poles you add *time constants* and invert. For *low*-frequency poles you add *frequencies* (in quadrature). Mixing the two is the most common arithmetic mistake in this topic.

5 Hybrid- π refresher

Nothing new here — this is the same model you used for one stage, restated so the symbols in §6 and §7 are unambiguous.

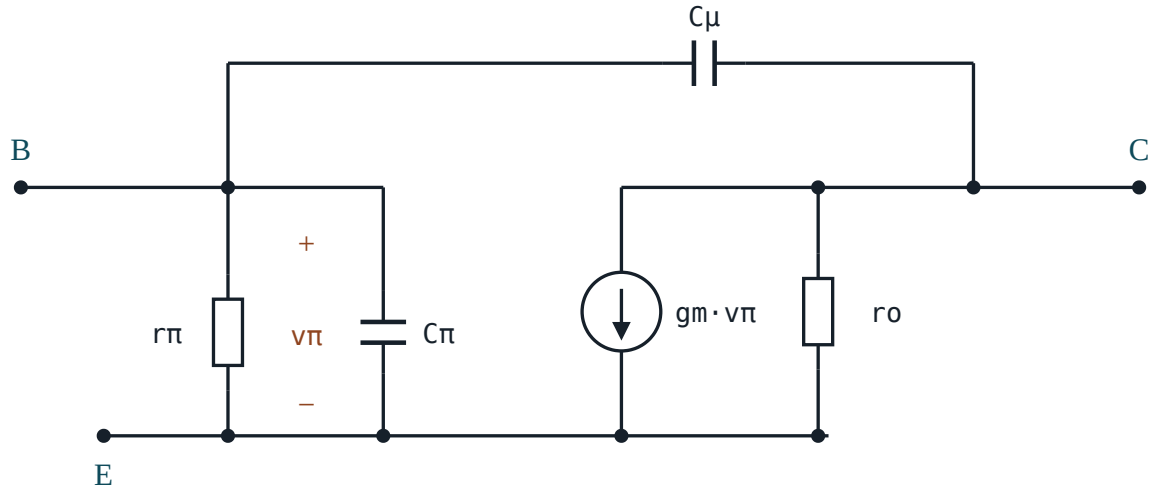


Figure 2 The hybrid- π model. C_μ is the troublemaker: it is the only element bridging input and output, so it is the only one that can be Miller-multiplied. Everything the cascode does is aimed at this one capacitor.

Parameters, with values at $I_C = 1$ mA, $\beta = 200$, $V_A = 100$ V, $f_T = 300$ MHz

Symbol	Definition	Depends on	Value
g_m	I_C/V_T	bias current only	38 mA/V
r_π	$\beta/g_m = \beta V_T/I_C$	current and β	5.3 k Ω
r_o	V_A/I_C	current and Early voltage	100 k Ω
C_π	$C_{je} + g_m \tau_F$	grows with I_C	15 pF
C_μ	C_{jc} (datasheet <code>C_obo</code>)	falls as V_{CB} rises	4 pF
f_T	$\frac{g_m}{2\pi(C_\pi + C_\mu)}$	the device's speed limit	300 MHz

Two results from Razavi Ch. 11 that we use constantly:

Miller input capacitance: $C_{in} = C_\mu (1 + |A_v|)$ **Miller output capacitance:** $C_{out} = C_\mu \left(1 + \frac{1}{|A_v|}\right)$

and, for a degenerated stage, the effective transconductance and the shrunken effective C_π :

$$G_m = \frac{g_m}{1 + g_m R_E}, \quad C_{\pi, \text{eff}} \approx \frac{C_\pi}{1 + g_m R_E}$$

Physical intuition – where Miller comes from

C_μ connects base to collector. When the base moves up by 1 mV, the collector moves down by $|A_v|$ mV. The voltage across C_μ therefore changes by $(1 + |A_v|)$ mV, so the charge the source must supply is $(1 + |A_v|)$ times larger than it would be for a capacitor to ground. The source cannot tell the difference between "a small capacitor whose far end runs away from me" and "a big capacitor sitting still." A 4 pF C_μ in a stage with gain 100 is a 404 pF load on the previous stage.

6 CE–CE cascade: small-signal analysis

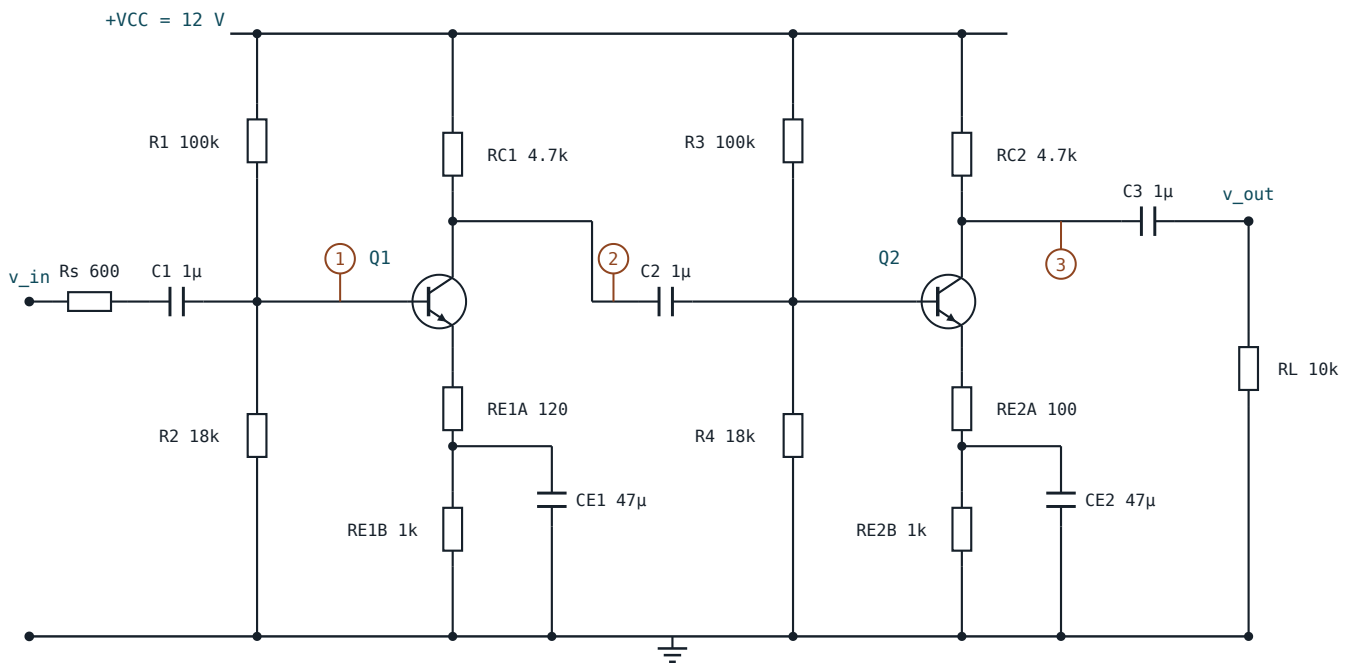


Figure 3 RC-coupled CE–CE cascade, with the component values designed in §8. Each emitter resistor is deliberately *split*: the lower part is bypassed for bias stability, the upper part is left unbypassed to set the gain accurately. The three circled nodes are the ones that carry all of the high-frequency capacitance.

6.1 Midband gain and impedances

At midband every coupling and bypass capacitor is a short and every device capacitance is an open. Working right to left, and writing $R_{B1} = R_1 \parallel R_2$, $R_{B2} = R_3 \parallel R_4$:

$$A_{v2} = - \frac{R_{C2} \parallel R_L \parallel r_{o2}}{1/g_{m2} + R_{E2A}} \quad R_{in2} = R_{B2} \parallel [r_{\pi2} + (\beta + 1)R_{E2A}]$$

$$A_{v1} = - \frac{R_{C1} \parallel r_{o1} \parallel R_{in2}}{1/g_{m1} + R_{E1A}} \quad R_{in} = R_{B1} \parallel [r_{\pi1} + (\beta + 1)R_{E1A}]$$

$$A_v = A_{v1}A_{v2} \text{ (positive: two inversions),} \quad R_{out} = R_{C2} \parallel r_{o2} \approx R_{C2}$$

Read the gain equation as a ratio of resistances

$|A_v| = R_{load, ac} / (1/g_m + R_E)$. The numerator is what the signal current flows *into*; the denominator is what it flows *out of*. The term $1/g_m$ is the transistor's own internal emitter resistance $r_e = V_T/I_E \approx 26 \Omega$ at 1 mA. Adding an external R_E of 120Ω therefore drops the gain by a factor of about 5.6 — but it makes that gain depend on a ratio of resistors instead of on I_C and temperature. That is a trade every designer makes on purpose.

6.2 High-frequency behaviour

Apply open-circuit time constants at the three circled nodes of Figure 3. Let $R'_S = R_S \parallel R_{B1}$ and let R_2^{node} be the total resistance at the interstage node.

Time constants of a CE–CE cascade

Node	Capacitance seen to ground	Resistance seen
1 — base of Q1	$C_{\pi1,eff} + C_{\mu1}(1 + A_{v1})$	$R_S \parallel R_{B1}$
2 — collector Q1 / base Q2	$C_{\mu1}\left(1 + \frac{1}{ A_{v1} }\right) + C_{\pi2,eff} + C_{\mu2}(1 + A_{v2})$	$R_{C1} \parallel r_{o1} \parallel R_{in2}$
3 — output	$C_{\mu2}\left(1 + \frac{1}{ A_{v2} }\right) + C_L$	$R_{C2} \parallel r_{o2} \parallel R_L$

Node 2 almost always dominates, for a reason that is easy to state: it is the only node that suffers a *large capacitance* (Miller-multiplied $C_{\mu2}$) and a *large resistance* (a few kilohms) at the same time. Node 1 has the same kind of capacitance but is driven from a low source resistance; node 3 has the same kind of resistance but only a small capacitance.

6.3 Low-frequency behaviour

Five capacitors, five high-pass corners. For each, find the resistance in series with it when the rest are shorted:

$$f_{C1} = \frac{1}{2\pi(R_S + R_{in})C_1}, \quad f_{C2} = \frac{1}{2\pi[(R_{C1} \parallel r_{o1}) + R_{in2}]C_2}, \quad f_{C3} = \frac{1}{2\pi[(R_{C2} \parallel r_{o2}) + R_L]C_3}$$

$$f_{CE} = \frac{1}{2\pi R_{E,eq}C_E}, \quad R_{E,eq} = R_{EB} \parallel \left[R_{EA} + \frac{r_\pi + R'_S}{\beta + 1} \right]$$

Physical intuition – why the bypass capacitor must be enormous

Looking up into the emitter you do not see r_π ; you see r_π divided by $(\beta + 1)$, because the emitter carries $(\beta + 1)$ times the base current. That is tens of ohms, not kilohms. The bypass capacitor must be low-impedance against *tens of ohms*, which is why it comes out at 47 μF or more while the coupling capacitors, which face kilohms, need only 1 μF . Nine times out of ten, an RC-coupled amplifier with a disappointing low-frequency response has an undersized emitter bypass capacitor.

7 The cascode amplifier

A cascode is a common-emitter stage whose collector, instead of driving a resistor directly, drives the *emitter of a second transistor* in common-base configuration. The name is old vacuum-tube shorthand for "cascade to cathode", and the arrangement is CE–CB.

Q1 converts input voltage to signal current. Q2 is a *current buffer*: it passes that current on to R_C almost unchanged ($i_c \approx \alpha i_e \approx i_e$) while presenting a very low resistance, about $1/g_{m2}$, at its emitter.

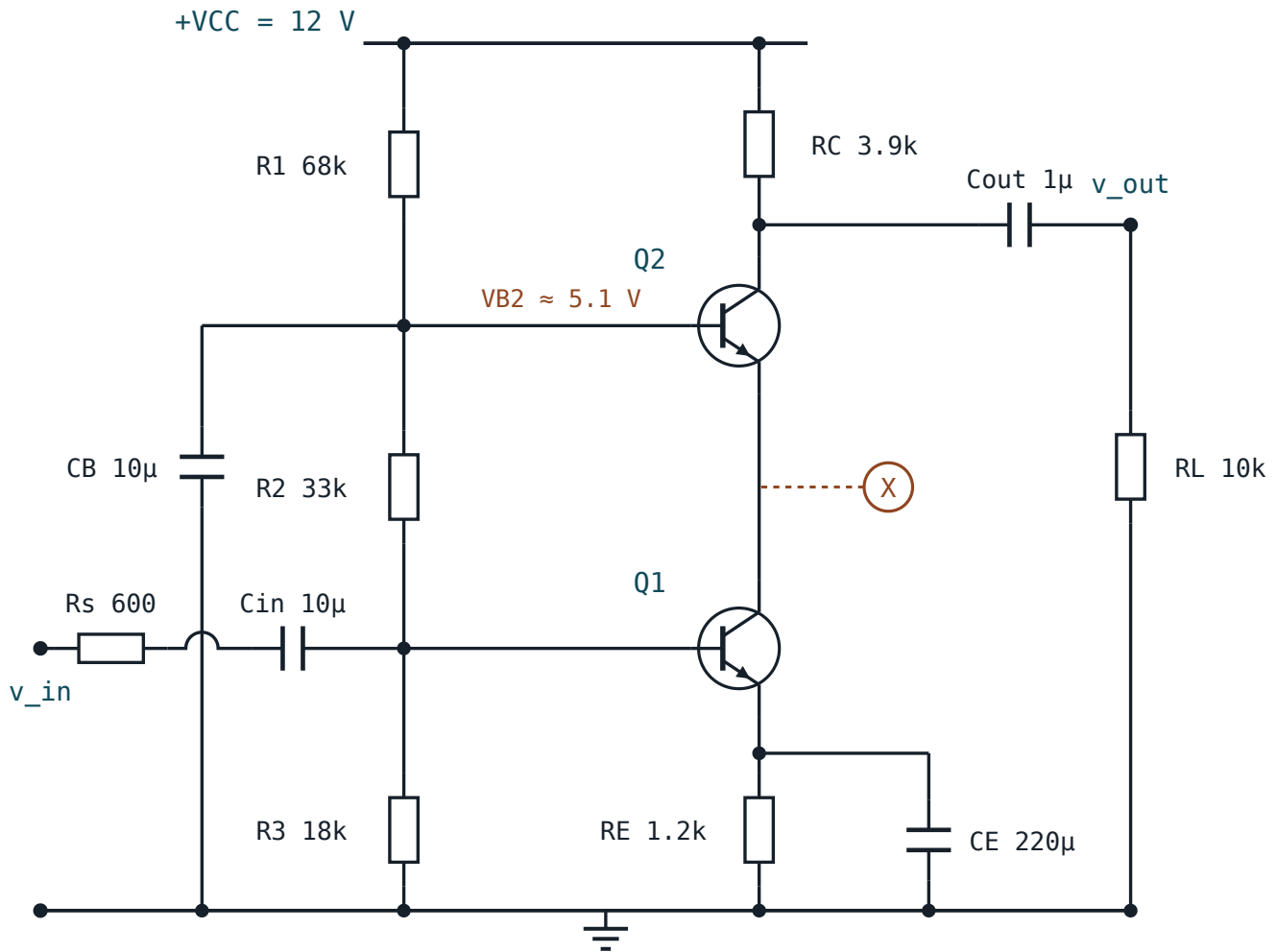


Figure 4 Cascode amplifier with the values designed in §9. The three-resistor divider sets both base voltages from one chain. C_B is not optional: without it, Q2's base is not an AC ground and the circuit degenerates into an ordinary CE stage with a mysteriously poor gain.

7.1 Why the Miller effect disappears

Look at what Q1's collector sees. It no longer sees R_C ; it sees the emitter of Q2, whose resistance is

$$R_X \approx \frac{1}{g_{m2}} = r_{e2} \approx 28 \Omega \quad \text{at } 1 \text{ mA}$$

So the voltage gain of the first transistor alone is

$$A_{v1} = -g_{m1}R_X \approx -\frac{g_{m1}}{g_{m2}} \approx -1$$

and the Miller multiplier collapses from $(1 + 100)$ to $(1 + 1) = 2$:

$$C_{in} = C_{\pi1} + C_{\mu1}(1 + 1) = 15 + 8 = 23 \text{ pF} \quad \text{instead of} \quad 15 + 4(1 + 100) = 419 \text{ pF}$$

Physical intuition – the brick wall at node X

Miller multiplication needs the collector to *swing*. Q2's emitter is a stiff, low-impedance node — whatever current Q1 pushes into it, the voltage there barely moves, because Q2 simply swallows the current and re-emits it at its collector. With no voltage swing at node X, there is no extra charge to shuttle through $C_{\mu 1}$, and the input node stops being slow.

Where does the voltage swing go? Up to Q2's collector, where it belongs — but Q2 is a common-base stage, and a CB stage has *no* capacitance bridging its input (emitter) and output (collector) other than the tiny C_{cs} . Its own $C_{\mu 2}$ runs from the collector to an AC ground, so it is just an ordinary load capacitance, not a Miller capacitance. **The cascode separates the node with high gain from the node with high impedance-to-ground. That separation is the entire trick.**

7.2 Midband gain

The signal current $g_{m1}v_{in}$ is delivered essentially intact to the collector load, so

$$A_v = -g_{m1}(R_C \parallel R_{out,casc} \parallel R_L) \approx -g_{m1}(R_C \parallel R_L)$$

with degeneration, replace g_{m1} by $G_m = g_{m1}/(1 + g_{m1}R_E)$. Note carefully: **with a resistive collector load, a cascode gives you no more gain than a plain CE stage**. Students often expect the gain to double because there are two transistors. It does not. What you buy is bandwidth, and output resistance.

7.3 Input and output resistance

$$R_{in} = R_B \parallel [r_{\pi 1} + (\beta + 1)R_E] \quad (\text{same as a CE stage})$$

The output resistance is where the cascode shines. Q1's r_{o1} appears as degeneration for Q2, and degeneration boosts output resistance:

$$R_{out,casc} = r_{o2} \left[1 + g_{m2}(r_{o1} \parallel r_{\pi 2}) \right] + (r_{o1} \parallel r_{\pi 2})$$

A BJT-only limitation worth knowing

In a MOSFET cascode there is no r_{π} , and R_{out} becomes $g_{m2}r_{o2}r_{o1}$ — the intrinsic gain times r_{o1} . In a BJT cascode, $r_{\pi 2} = \beta/g_{m2}$ is typically *much smaller* than r_{o1} (5.5 k Ω versus 100 k Ω), so the parallel combination is pinned at $r_{\pi 2}$ and

$$R_{out,casc} \approx g_{m2}r_{o2}r_{\pi 2} = \beta_2 r_{o2}$$

The bipolar cascode output resistance **saturates at βr_o** — here about 20 M Ω . Stacking a third transistor buys you almost nothing. This is exactly the argument Razavi makes in Ch. 9, and it is a favourite viva question.

7.4 High-frequency behaviour

Time constants of a cascode

Node	Capacitance	Resistance	Comment
Base of Q1	$C_{\pi 1} + 2C_{\mu 1}$	$R_S \parallel R_B$	small, because Miller is gone
Node X	$C_{\mu 1} + C_{\pi 2}$	$\approx 1/g_{m2}$	negligible — tens of ohms
Output	$C_{\mu 2} + C_L$	$R_C \parallel R_L$	now the dominant pole

The bandwidth improvement therefore grows with the source resistance. If your source is a $50\ \Omega$ generator, the input node was never the problem and the cascode helps only modestly. If your source is a $10\ \text{k}\Omega$ sensor, the cascode can improve bandwidth by an order of magnitude. In §9 we quantify both cases.

7.5 What the cascode costs

- **Headroom.** Two $V_{CE,sat}$ stack up, and Q2's emitter sits at $V_{B2} - 0.7$. Output swing shrinks by roughly 1–1.5 V compared with a single CE stage on the same supply.
- **An extra bias node** that must be both correct in DC and dead-quiet in AC.
- **Slightly more noise** and one more device.
- **No gain benefit at all** unless the load is a current source rather than a resistor. In an IC, where R_C can be a current source of $\sim \beta r_o$, the cascode's high R_{out} turns into a gain of $\sim g_m \beta r_o$. On a breadboard with a $3.9\ \text{k}\Omega$ resistor, it does not.

8 Design 1 — CE–CE cascade for $A_v = 500$

Specification. $A_v = v_{out}/v_{in} = 500$ (54 dB) at midband, $V_{CC} = 12\ \text{V}$, $R_S = 600\ \Omega$, $R_L = 10\ \text{k}\Omega$, $f_L \leq 100\ \text{Hz}$. Devices: BC547B or 2N2222 ($\beta \approx 200$, $V_A \approx 100\ \text{V}$, $f_T \approx 300\ \text{MHz}$).

- S1 Choose the bias current.** Take $I_C = 1\ \text{mA}$ for both stages — low enough that R_C can be a few kilohms, high enough that f_T is near its peak and r_π is not uselessly small. At this current $g_m \approx 38\ \text{mA/V}$, $r_\pi \approx 5.3\ \text{k}\Omega$, $r_o \approx 100\ \text{k}\Omega$.
- S2 Set the DC operating point.** Put about $0.1V_{CC}$ across the emitter resistor for thermal stability, and leave V_{CE} comfortably above the swing you need: $V_E \approx 1.1\ \text{V}$ so $R_{EA} + R_{EB} \approx 1.1\ \text{k}\Omega$; $R_C = 4.7\ \text{k}\Omega$ giving $V_{RC} = 4.4\ \text{V}$ and $V_{CE} \approx 6.5\ \text{V}$.

- S3 Design the divider.** Rule of thumb: divider current $\geq 10I_B = 50 \mu\text{A}$, so $R_1 + R_2 \approx 120 \text{ k}\Omega$. For $V_B = V_E + 0.7 \approx 1.8 \text{ V}$ take $R_1 = 100 \text{ k}\Omega$, $R_2 = 18 \text{ k}\Omega$ ($V_{Th} = 1.83 \text{ V}$, $R_{Th} = 15.25 \text{ k}\Omega$). Verify with base-current loading included:

$$I_E = \frac{V_{Th} - V_{BE}}{R_E + R_{Th}/(\beta + 1)} = \frac{1.83 - 0.7}{1120 + 76} = 0.95 \text{ mA} \checkmark$$

- S4 Split the gain between the stages.** Two undegenerated CE stages here would give about 8000 — far more than asked. We must *throw gain away deliberately*, and the right place to throw it is unbypassed emitter resistance, because that also makes the gain insensitive to β , temperature and device spread. Give stage 2 slightly more than stage 1 (the later stage is less heavily loaded): target roughly 21×24 .

- S5 Size R_{E2A} , starting at the output and working backwards.**

$$R_{L2} = R_{C2} \parallel R_L \parallel r_{o2} = 4.7 \parallel 10 \parallel 106 = 3.10 \text{ k}\Omega$$

$$A_{v2} = \frac{3100}{1/g_m + R_{E2A}} = 24 \Rightarrow R_{E2A} \approx 100 \Omega \Rightarrow A_{v2} = \frac{3100}{27.8 + 100} = 24.3$$

- S6 Compute stage 2's input resistance — this is stage 1's load.**

$$R_{in2} = R_{B2} \parallel [r_{\pi2} + (\beta + 1)R_{E2A}] = 15.25 \parallel (5.53 + 20.1) = 15.25 \parallel 25.6 = 9.56 \text{ k}\Omega$$

Notice the damage. Stage 1's collector was going to see $4.7 \text{ k}\Omega$; it now sees $4.7 \parallel 106 \parallel 9.56 = 3.06 \text{ k}\Omega$.

- S7 Size R_{E1A} for the gain that remains.**

$$A_{v1}^{\text{req}} = \frac{500}{24.3} = 20.6 \Rightarrow \frac{1}{g_m} + R_{E1A} = \frac{3060}{20.6} = 148.5 \Omega \Rightarrow R_{E1A} = 121 \Omega \rightarrow 120 \Omega$$

Then $A_{v1} = 3060/147.8 = 20.7$ and $A_v = 20.7 \times 24.3 = 503$ — 0.6 % high, well inside component tolerance. Set $R_{E1B} = 1 \text{ k}\Omega$ so the total emitter resistance stays at its biased value.

- S8 Size the capacitors for f_L .** Find the resistance each one faces and aim for every individual corner below about 30 Hz.

$$R_{in} = 15.25 \parallel (5.53 + 24.1) = 10.1 \text{ k}\Omega \Rightarrow C_1 : R = 600 + 10100 = 10.7 \text{ k}\Omega \Rightarrow 1 \mu\text{F} \rightarrow 14.9 \text{ Hz}$$

$$C_2 : R = 4.5 \text{ k} + 9.56 \text{ k} = 14.1 \text{ k}\Omega \Rightarrow 1 \mu\text{F} \rightarrow 11.3 \text{ Hz} \quad C_3 : R = 4.5 \text{ k} + 10 \text{ k} \Rightarrow 1 \mu\text{F} \rightarrow 11.0 \text{ Hz}$$

$$C_E : R_{E,eq} = 1000 \parallel \left[120 + \frac{5530 + 577}{201} \right] = 131 \Omega \Rightarrow 47 \mu\text{F} \rightarrow 25.9 \text{ Hz}$$

Root-sum-square gives $f_L \approx \sqrt{14.9^2 + 11.3^2 + 11.0^2 + 25.9^2 + 26.8^2} \approx 43 \text{ Hz}$, comfortably inside the 100 Hz specification.

- S9 Predict f_H with open-circuit time constants, using $C_\pi = 15 \text{ pF}$, $C_\mu = 4 \text{ pF}$ and 20 pF of probe-plus-wiring capacitance at the output.**

Node	C (pF)	R (Ω)	τ (ns)	share
1 — base of Q1	89.6	577	51.7	11 %
2 — interstage	108.7	3060	332.6	72 %
3 — output	24.2	3103	75.0	16 %
Total			459	

$$f_H \approx \frac{1}{2\pi \times 459 \text{ ns}} \approx \mathbf{347 \text{ kHz}}$$

Nearly three-quarters of the delay sits at the interstage node, and almost all of that is $C_{\mu 2}(1 + 24.3) = 101 \text{ pF}$ of Miller capacitance from stage 2.

Design 1 – bill of materials

Component	Value	Component	Value
R1, R3 (upper divider)	100 k Ω	C1, C2, C3 (coupling)	1 μF
R2, R4 (lower divider)	18 k Ω	CE1, CE2 (bypass)	47 μF , 16 V
RC1, RC2	4.7 k Ω	Q1, Q2	BC547B or 2N2222
RE1A / RE2A (unbypassed)	120 Ω / 100 Ω	V_{CC}	12 V
RE1B, RE2B (bypassed)	1 k Ω	I_C per stage	$\approx 0.95 \text{ mA}$

Design 1 – predicted performance

Quantity	Predicted	Where it comes from
A_{v1}	-20.7	loaded by R_{in2}
A_{v2}	-24.3	loaded by R_L
$A_v = v_{out}/v_{in}$	+503 (54.0 dB)	non-inverting — two inversions
$A_{vs} = v_{out}/v_s$	+475 (53.5 dB)	after the R_S – R_{in} divider
R_{in}	10.1 k Ω	raised by R_{E1A}
R_{out}	4.5 k Ω	$R_{C2} \parallel r_{o2}$
f_L	$\approx$ 43 Hz	emitter bypass dominates
f_H	$\approx$ 347 kHz	interstage node dominates
Gain–bandwidth product	$\approx$ 174 MHz	vs. 51 MHz for one stage of the same type

Before you power it up

At a gain of 500, an input of 10 mV_{pk} demands 5 V_{pk} out — more than the available swing, so you will measure a clipped waveform and conclude the design is wrong. Set the generator so the amplifier input is about **2 mV_{pk}** (2 mV in → 1 V out). Most bench generators cannot deliver that cleanly, so build a 100:1 attenuator at the input from a 10 k Ω and a 100 Ω resistor and drive it with 200 mV. Probe the *attenuated* node with channel 1, never the generator terminals.

9 Design 2 — cascode for $A_v = 100$

Specification. $A_v = 100$ (40 dB), $V_{CC} = 12$ V, $R_S = 600$ Ω , $R_L = 10$ k Ω , bandwidth as wide as possible, same transistors.

The gain here is modest, so no degeneration is needed at signal frequencies — R_E is fully bypassed and exists purely for DC stability. The design problem is really a *headroom budgeting* problem, because the cascode stacks two collector–emitter drops between the supply rails.

S1 Budget the supply voltage before choosing any resistor. Working up from ground:

$$V_E + V_{CE1} + V_{CE2} + V_{R_C} = 1.05 + 3.3 + 3.9 + 3.7 = 12 \text{ V}$$

V_E buys bias stability, V_{CE1} must stay above about 1 V, V_{CE2} sets how far the output can swing downwards, and V_{R_C} sets the gain. Compare this with the single CE design, where all 11 V were available to split between V_{CE} and V_{R_C} . That extra V_{CE} is the price of the cascode.

S2 Pick the current and the bottom of the chain. $I_C \approx 0.9 \text{ mA}$, $R_E = 1.2 \text{ k}\Omega$ giving $V_E \approx 1.05 \text{ V}$ and $V_{B1} \approx 1.75 \text{ V}$.

S3 Choose V_{B2} . Q2's emitter sits at $V_{B2} - 0.7$, and that node is the floor of the output swing. Putting V_{B2} near $0.4V_{CC}$ shares the headroom evenly:
 $V_{B2} \approx 5.0 \text{ V} \Rightarrow V_{E2} = V_{C1} \approx 4.3 \text{ V}$, so $V_{CE1} = 3.3 \text{ V}$.

S4 Design the three-resistor chain for a current of roughly $100 \mu\text{A}$ (comfortably more than the two base currents of about $4.5 \mu\text{A}$ each):

$$R_1 : R_2 : R_3 = (12 - 5.0) : (5.0 - 1.75) : 1.75 \Rightarrow R_1 = 68 \text{ k}\Omega, R_2 = 33 \text{ k}\Omega, R_3 = 18 \text{ k}\Omega$$

Checking with base-current loading: $V_{B1} = 1.75 \text{ V}$, $V_{B2} = 5.02 \text{ V}$,

$$I_E = \frac{1.815 - 0.7}{1200 + 15280/201} = 0.874 \text{ mA} \Rightarrow g_m = 33.5 \text{ mA/V}, r_\pi = 5.97 \text{ k}\Omega, r_o = 115 \text{ k}\Omega$$

S5 Set the gain with R_C . Because the cascode's own output resistance ($\approx \beta r_{o2} = 23 \text{ M}\Omega$) is thousands of times larger than R_C , it drops out of the parallel combination entirely:

$$|A_v| = g_{m1} (R_C \parallel R_L) = 100 \Rightarrow R_C \parallel R_L = 2.99 \text{ k}\Omega \Rightarrow R_C = 4.3 \text{ k}\Omega$$

$$|A_v| = 33.5 \text{ m} \times (4.3 \parallel 10) = 33.5 \text{ m} \times 3.007 \text{ k} = 100.7 \text{ (40.1 dB)}$$

Confirm the DC point: $V_{C2} = 12 - 0.87 \text{ m} \times 4.3 \text{ k} = 8.26 \text{ V}$, so $V_{CE2} = 3.94 \text{ V}$ and the output can swing about $\pm 3.6 \text{ V}$. Good.

S6 Size C_B — the one that students forget. Q2's base must be a *hard* AC ground. It faces $R_1 \parallel (R_2 + R_3) = 29 \text{ k}\Omega$, so $10 \mu\text{F}$ puts its corner at 0.55 Hz . Undersize this capacitor and Q2 stops behaving like a common-base stage; node X regains its swing, Miller multiplication returns, and you have built an ordinary CE amplifier with two transistors in it.

S7 Size the remaining capacitors.

$$R_{in} = R_B \parallel r_{\pi1} = 15.28 \parallel 5.97 = 4.29 \text{ k}\Omega$$

$$C_{in} : R = 600 + 4290 = 4.89 \text{ k}\Omega \Rightarrow 10 \mu\text{F} \rightarrow 3.3 \text{ Hz} \quad C_{out} : R = 4.3 \text{ k} + 10 \text{ k} \Rightarrow 1 \mu\text{F} \rightarrow 11.1 \text{ Hz}$$

$$C_E : R_{E,eq} = 1200 \parallel \frac{r_\pi + R'_S}{\beta + 1} = 1200 \parallel 32.6 = 31.7 \Omega \Rightarrow 220 \mu\text{F} \rightarrow 22.8 \text{ Hz}$$

Overall $f_L \approx 26 \text{ Hz}$. Note that the emitter bypass capacitor is now $220 \mu\text{F}$, nearly five times the value used in Design 1, purely because R_E is fully bypassed and the resistance looking into the emitter is only $\approx r_e = 30 \Omega$.

S8 Predict f_H . With $C_\pi = 14 \text{ pF}$, $C_\mu = 4 \text{ pF}$, $C_L = 20 \text{ pF}$:

Node	C (pF)	R (Ω)	τ (ns)	share
Base of Q1	22	577	12.7	15 %
Node X	18	30	0.5	< 1 %
Output	24	3007	72.2	85 %
Total			85.4	

$$f_H \approx \frac{1}{2\pi \times 85.4 \text{ ns}} \approx 1.86 \text{ MHz}$$

The picture has inverted. In the CE–CE cascade the input side dominated; here it contributes 15 %, and the amplifier is limited only by the load it has to drive.

9.1 The comparison that makes the point

Build the same 40 dB stage *without* the cascode transistor — identical bias, identical R_C , R_E fully bypassed — and the input node's capacitance becomes

$C_\pi + C_\mu(1 + 100.7) = 14 + 407 = 421 \text{ pF}$. Its time constant alone is 243 ns, and

$$f_H^{(\text{plain CE})} = \frac{1}{2\pi(243 + 72) \text{ ns}} \approx 505 \text{ kHz}$$

Same gain, same devices, same supply – only the topology differs

Source resistance	Plain CE	Cascode	Improvement
$R_S = 600 \Omega$ (bench generator)	505 kHz	1.86 MHz	3.7 ×
$R_S = 10 \text{ k}\Omega$ (sensor, previous stage)	61 kHz	773 kHz	12.7 ×

Physical intuition – when the cascode is worth it

The cascode removes a *capacitance* at the input node. Its benefit therefore scales with the *resistance* that capacitance sits against — that is, with the source resistance. Drive the stage from a stiff $50\ \Omega$ generator and the input pole was never in the way; the cascode buys you little. Drive it from a high-impedance source and the cascode is transformative. This is exactly why cascodes appear everywhere in IC design, where every stage is driven by the high output resistance of the stage before it, and only occasionally in discrete bench circuits.

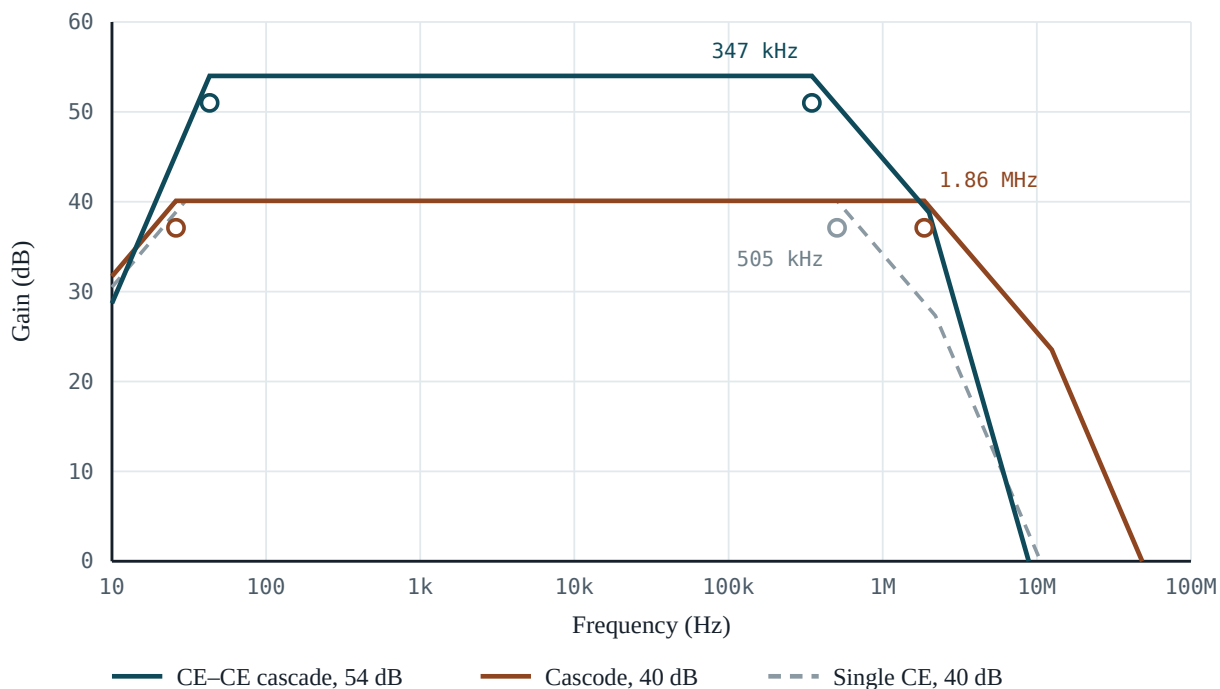


Figure 5 Asymptotic magnitude response of the three circuits, predicted by the open-circuit time-constant estimates. Circles mark the -3 dB points. Read it as two separate stories: the cascade *buys 14 dB of gain and pays with bandwidth*; the cascode *keeps the same gain and gets the bandwidth back*. Your measured curves will round off at the corners rather than break sharply, and the roll-off will begin a little earlier, because the OCTC method is deliberately pessimistic.

Summary of both laboratory designs

Quantity	Single CE	CE–CE (Expt 1)	Cascode (Expt 2)
Midband gain	101 (40.1 dB)	503 (54.0 dB)	101 (40.1 dB)
Phase at midband	180°	0°	180°
f_L	≈ 30 Hz	≈ 43 Hz	≈ 26 Hz
f_H	505 kHz	347 kHz	1.86 MHz
Gain–bandwidth	51 MHz	174 MHz	188 MHz
R_{in}	4.3 k Ω	10.1 k Ω	4.3 k Ω
R_{out}	4.3 k Ω	4.5 k Ω	4.3 k Ω
Transistors	1	2	2

10 Laboratory procedure

10.1 Bring-up, before any signal

1. Build the DC circuit only — resistors and transistors, no capacitors, no generator.
2. Measure V_B , V_E and V_C at every transistor and compare with the design. If V_C is within 0.2 V of V_E the device is saturated; if $V_C \approx V_{CC}$ it is cut off. Either way, stop and fix the bias before going further — a saturated transistor has almost no gain and no amount of signal adjustment will rescue it.
3. Compute $I_C = V_{RC}/R_C$ from your measurement and recompute $g_m = I_C/26\text{mV}$. Use this **measured g_m in your report's predicted gain**, not the design value; a ± 20 % spread in β between two BC547s from the same bag is entirely normal.
4. Now add the capacitors, observing electrolytic polarity: the positive terminal of a coupling capacitor goes toward the higher DC potential; for the emitter bypass, positive goes to the emitter.

10.2 Midband gain

1. Set the generator to 1 kHz — safely inside the midband for both designs.
2. Start with a tiny input and increase it while watching the output. Find the level at which the output *just* begins to flatten, then back off to about half of it. Record v_{in} and v_{out} at that level.

3. $A_v = v_{out}/v_{in}$, both measured peak-to-peak on the scope. Note the phase: 0° for the CE–CE cascade, 180° for the cascode.
4. For the cascade, also measure the interstage node so you can report A_{v1} and A_{v2} separately. Their product should equal the overall gain — if it does not, you have a loading or bypass problem in one stage.

10.3 Frequency response

1. Hold the *input* amplitude constant. Sweep frequency logarithmically: 10, 20, 50, 100, 200, 500 Hz, then the same 1–2–5 pattern per decade up to 10 MHz. Verify the input amplitude at each point — generator output drops at high frequency into a low impedance.
2. Record v_{out} and compute $20 \log_{10}(v_{out}/v_{in})$.
3. Plot dB against frequency on semilog paper (or a log x-axis in your spreadsheet). Read f_L and f_H where the curve falls 3 dB below the flat midband value, and report $BW = f_H - f_L$ and the gain–bandwidth product.
4. Take extra points near each corner — five points per decade elsewhere, ten near the knees.

10.4 Input and output resistance

Both are two-measurement substitution tests at 1 kHz.

$$R_{in} = R_{test} \frac{v_2}{v_1 - v_2} \quad (v_1 = \text{amplifier input voltage with } R_{test} \text{ shorted, } v_2 \text{ with it inserted})$$

$$R_{out} = R_L \left(\frac{v_{oc}}{v_L} - 1 \right) \quad (v_{oc} = \text{output with } R_L \text{ removed, } v_L = \text{with it connected})$$

Choose R_{test} comparable to the expected R_{in} — 10 k Ω works for both designs.

Five ways this experiment goes wrong

1. **$\times 1$ probe.** A $\times 1$ probe adds roughly 100 pF at the output node. In Design 1 that turns 24 pF into 124 pF and drags f_H down by a factor of about 1.8. **Use $\times 10$ probes and compensate them** against the scope's square-wave reference before you start.
2. **Clipping mistaken for roll-off.** If the gain seems to fall at *all* frequencies, the input is too large. Halve it and see whether the measured gain rises.
3. **Undersized bypass capacitor.** A gain that is far below prediction but flat with frequency usually means C_E is not doing its job. Temporarily parallel another 100 μF across it — if the gain jumps, that was the cause.
4. **Missing or undersized C_B in the cascode.** Same symptom, different node. Check for signal at Q2's base with the scope: it should be flat zero.
5. **Long leads and oscillation.** A cascade with a gain of 500 will happily oscillate on a breadboard. Keep the input wiring short and away from the output, put a 0.1 μF ceramic capacitor from the V_{CC} rail to ground next to each stage, and use one common ground point rather than a chain of ground jumpers.

10.5 Simulate first

Run both circuits in LTspice or NgSpice before you touch the breadboard, so you know what shape you are looking for. The idealised model below is tuned to match the hand calculations; substituting the manufacturer's 2N2222 or BC547 model afterwards is a useful reality check.

```

* ---- Experiment 1: CE-CE cascade, Av = 500 ----
VCC 100 0 DC 12
Vs 1 0 AC 1 SIN(0 2m 1k)
Rs 1 2 600
C1 2 3 1u
R1 100 3 100k
R2 3 0 18k
Q1 4 3 5 QN
RC1 100 4 4.7k
RE1A 5 6 120
RE1B 6 0 1k
CE1 6 0 47u
C2 4 7 1u
R3 100 7 100k
R4 7 0 18k
Q2 8 7 9 QN
RC2 100 8 4.7k
RE2A 9 10 100
RE2B 10 0 1k
CE2 10 0 47u
C3 8 11 1u
RL 11 0 10k
CL 11 0 20p ; scope probe + wiring
.model QN NPN(Is=1e-14 Bf=200 Vaf=100 Tf=0.32n Cje=4p Cjc=4p Rb=10 Rc=
.op
.ac dec 100 1 100Meg
.end

```

```

* ---- Experiment 2: Cascode, Av = 100 ----
VCC 100 0 DC 12
Vs 1 0 AC 1 SIN(0 10m 1k)
Rs 1 2 600
Cin 2 3 10u
R1 100 4 68k
R2 4 3 33k
R3 3 0 18k
CB 4 0 10u ; Q2 base AC ground - do not omit
Q1 5 3 6 QN
Q2 7 4 5 QN
RC 100 7 4.3k
RE 6 0 1.2k
CE 6 0 220u
Cout 7 8 1u
RL 8 0 10k
CL 8 0 20p
.model QN NPN(Is=1e-14 Bf=200 Vaf=100 Tf=0.32n Cje=4p Cjc=4p Rb=10 Rc=
.op
.ac dec 100 1 100Meg
.end

```

Plot **V(11)** and **V(8)** in dB. Two experiments worth running in the simulator because they are awkward on the bench: short **CB** out of Design 2 and watch the bandwidth collapse toward the plain-CE curve; and sweep **Rs** from 100 Ω to 100 k Ω in a **.step** to see the cascode's advantage grow.

11 Formula sheet

Device

$$g_m = \frac{I_C}{V_T}, \quad r_\pi = \frac{\beta}{g_m}, \quad r_o = \frac{V_A}{I_C}, \quad r_e = \frac{V_T}{I_E} \approx \frac{1}{g_m}, \quad g_m r_o = \frac{V_A}{V_T}, \quad f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

Degenerated CE stage

$$A_v = -\frac{R_{L,ac}}{1/g_m + R_E}, \quad R_{in} = R_B \parallel [r_\pi + (\beta+1)R_E], \quad R_{out} = R_C \parallel r_o$$

Cascading

$$A_v = \prod_k A_{vk} \text{ loaded}, \quad A_{vs} = \frac{R_{in}}{R_{in} + R_s} A_v, \quad f_H^{(n)} = f_H \sqrt{2^{1/n} - 1}, \quad f_L^{(n)} = \frac{f_L}{\sqrt{2^{1/n} - 1}}$$

Frequency response

$$C_{in} = C_\mu(1 + |A_v|), \quad C_{out} = C_\mu\left(1 + \frac{1}{|A_v|}\right), \quad f_H \approx \frac{1}{2\pi \sum_i R_{i0}C_i}, \quad f_L \approx \sqrt{\sum_i f_{Li}^2}$$

Cascode

$$A_v \approx -g_{m1}(R_C \parallel R_L), \quad R_X \approx \frac{1}{g_{m2}}, \quad R_{out} = r_{o2}[1 + g_{m2}(r_{o1} \parallel r_{\pi2})] + (r_{o1} \parallel r_{\pi2}) \approx \beta_2 r_{o2}$$

Bench measurements

$$R_{in} = R_{test} \frac{v_2}{v_1 - v_2}, \quad R_{out} = R_L \left(\frac{v_{oc}}{v_L} - 1 \right), \quad A_{dB} = 20 \log_{10} \frac{v_{out}}{v_{in}}$$

12 Exercises

1. In Design 1, R_{E1A} is changed from 120 Ω to 220 Ω and nothing else. Find the new A_v , R_{in} and f_H . Explain why the bandwidth *improves* even though only a resistor changed.
2. Insert an emitter follower between the two stages of Design 1, biased at 1 mA. Recompute A_{v1} , the overall gain and the dominant time constant. How much bandwidth did the buffer buy, and at what cost in gain?
3. A three-stage amplifier is built from identical stages, each with gain 12 and $f_H = 900$ kHz. Assuming no interaction, find the overall gain in dB and the overall f_H . Then explain, in two sentences, why the real circuit will fall well short of your answer.
4. Show that the CE gain expression $|A_v| = V_{RC}/V_T$ follows from $g_m R_C$, and use it to find the maximum gain of a single undegenerated CE stage on a 5 V supply that must deliver 1 V_{pk} at the output.
5. For Design 2, compute R_{out} using the full cascode expression and then using βr_{o2} . How close are they, and what would R_{out} be if the devices were MOSFETs with the same g_m and r_o ?
6. In Design 2, C_B is accidentally left out. Sketch the resulting small-signal circuit at Q2's base, argue what the gain of Q1 into node X becomes, and estimate the new f_H .
7. Design a CE–CE cascade for $A_v = 200$ on a 9 V supply driving a 4.7 k Ω load, with $f_L \leq 50$ Hz. Give a full bill of materials with standard E12 values and predict f_H .
8. An amplifier must deliver a gain of 60 dB with the widest possible bandwidth from devices with $f_T = 300$ MHz. Would you use two stages of gain 32 or three stages of gain 10? Justify with numbers, and state one non-bandwidth reason that might override your answer.

Prepared by **Ameenudeen P E** for the circuit theory course and its companion laboratory. Notation, the hybrid- π model, the Miller approximation and the open-circuit time-constant method follow B. Razavi, *Fundamentals of Microelectronics*, 2nd ed., Chapters 5, 9 and 11; read §9.1 and §11.4 alongside §7 of this note. All numerical results assume $V_T = 26 \text{ mV}$, $\beta = 200$, $V_A = 100 \text{ V}$ and $f_T = 300 \text{ MHz}$; expect measured values to differ by 10–20 % with real devices.